

# Complex Analysis

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# 1 Complex Number System

## Definition 1.1

## Complex Numbers

The set of complex numbers  $\mathbb{C}$  is  $\mathbb{R}^2$ , together with the operations

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

and

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1)$$

## Theorem 1.1

## Field Structure

$\mathbb{C}$  is a field. With  $0 = (0, 0)$ ,  $-(x, y) = (-x, -y)$ ,  $1 = (1, 0)$  and

$$(x, y)^{-1} = \left( \frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right) \text{ for } (x, y) \neq (0, 0)$$

## Notation 1.1

If  $z = (x, y)$  we call  $x$  the “real part” of  $z$  and write  $x = \operatorname{Re}(z)$ .

We call  $y$  the “imaginary part” of  $z$  and write  $y = \operatorname{Im}(z)$ .

Define  $i = (0, 1)$ .

## Remark 1.2

There is an embedding

$$\begin{aligned} \mathbb{R} &\hookrightarrow \mathbb{C} \\ x &\mapsto (x, 0) \end{aligned}$$

so we view  $\mathbb{R} \subseteq \mathbb{C}$ .

## Notation 1.3

Under the above embedding, any  $z \in \mathbb{C}$  can be written as  $z = \operatorname{Re}(z) + \operatorname{Im}(z) \cdot i$

and  $i^2 = -1$

## Definition 1.2

## Conjugation

Define *conjugation* to be the map

$$\begin{aligned} - : \mathbb{C} &\longrightarrow \mathbb{C} \\ z = x + yi &\longmapsto x - yi = \bar{z} \end{aligned}$$

**Note**

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2}, \quad \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$$

**Note**

If  $z = x + yi \in \mathbb{C}$  it has a polar representation  $(x, y) \mapsto (r, \theta)$  via  $z = r \cos \theta + ir \sin \theta$  where  $r = \sqrt{x^2 + y^2} \geq 0$ ,  $x = r \cos \theta$ ,  $y = r \sin \theta$ . The angle  $\theta$  is defined modulo  $2\pi$  when  $z \neq 0$ , and is arbitrary when  $z = 0$ .

**Remark 1.4**

If  $z_1, z_2$  have polar representations

$$z_1 : (r_1, \theta_1), \quad z_2 : (r_2, \theta_2)$$

Then

$$\begin{aligned} z_1 \cdot z_2 &= (r_1 \cos \theta_1 + ir_1 \sin \theta_1) \cdot (r_2 \cos \theta_2 + ir_2 \sin \theta_2) \\ &= r_1 r_2 ((\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2)) \\ &= r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)) \end{aligned}$$

**Note**

If  $z = x + iy$  and  $r = \sqrt{x^2 + y^2}$ , then  $r^2 = z\bar{z}$ .

**Remark 1.5**

$\mathbb{C}$  is a metric space and thus the triangle inequality holds.

**Remark 1.6**

$\mathbb{C}$  is a topological space, so topological notions make sense.

**Definition 1.3****Argument**

If  $z \in \mathbb{C} \setminus \{0\}$  has polar representation  $(r, \theta)$ , we call  $\theta$  the *argument* of  $z$  and write  $\theta = \arg(z)$ . We'll think about  $\arg$  as a multivalued function. We write  $\operatorname{Arg}(z) = \theta$  if  $\theta \in (-\pi, \pi]$ , the *principal value*.

**Example 1.1**

If  $z = r(\cos \theta + i \sin \theta)$  and  $z \neq 0$  then we can find an  $n^{\text{th}}$  root of  $z$  (actually  $n$  of them).

## 2 Differentiation

### Definition 2.1

### Complex Derivative

Suppose  $f : U \rightarrow \mathbb{C}$  where  $U \subseteq \mathbb{C}$  is open and contains  $z_0$ . We'll say that  $f$  has derivative  $f'(z_0)$  at  $z_0$  if

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

### Theorem 2.1

### Product Rule

If  $f$  and  $g$  are differentiable at  $z_0$  then so is  $fg$  and

$$(f \cdot g)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0)$$

### Remark 2.1

If  $f : \mathbb{C} \rightarrow \mathbb{C}$ , then we may write  $f(z) = u(z) + iv(z)$  where  $u, v : \mathbb{C} \rightarrow \mathbb{R}$ . If  $f$  is differentiable at  $z_0 = x_0 + iy_0$  (say  $f'(z_0) = a + ib$ ), then

$$\begin{aligned} R(z) &= f(z) - f(z_0) - f'(z_0)(z - z_0) \\ &= u(z) - u(z_0) - a(x - x_0) + b(y - y_0) + i(v(z) - v(z_0) - b(x - x_0) - a(y - y_0)) \\ &= R_1(z) + iR_2(z) \end{aligned}$$

Then  $\lim_{z \rightarrow z_0} R_1 \frac{z}{|z - z_0|} = \lim_{z \rightarrow z_0} R_2 \frac{z}{|z - z_0|} = 0$ . Thus both  $u$  and  $v$  are differentiable and  $\frac{\partial u}{\partial x}(z_0) = a$ ,  $\frac{\partial u}{\partial y}(z_0) = -b$ ,  $\frac{\partial v}{\partial x}(z_0) = b$ ,  $\frac{\partial v}{\partial y}(z_0) = a$ .

### Theorem 2.2

### Cauchy-Riemann Equations

If  $f : \mathbb{C} \rightarrow \mathbb{C}$ ,  $f(z) = u(z) + iv(z)$ , then  $f$  is differentiable at  $z_0$  (as a complex function) iff both  $u$  and  $v$  are differentiable at  $z_0$  and

$$\begin{aligned} \frac{\partial u}{\partial x}(z_0) &= \frac{\partial v}{\partial y}(z_0) \quad \text{and} \\ \frac{\partial v}{\partial x}(z_0) &= -\frac{\partial u}{\partial y}(z_0) \end{aligned}$$

These are called the Cauchy-Riemann equations.

### Note

If  $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ , then  $u$  is differentiable at  $(x_0, y_0)$  if  $\frac{\partial u}{\partial x}$  and  $\frac{\partial u}{\partial y}$  exist and are continuous in neighborhoods of  $(x_0, y_0)$ .

**Definition 2.2****Exponential function**

The exponential function is defined to be the function

$$\exp(x + iy) = e^x(\cos y + i \sin y)$$

we write  $e^z := \exp(z)$ .

**Remark 2.2**

For  $y \in \mathbb{R}$ , we have  $e^{iy} = \cos(y) + i \sin(y)$ .

**Definition 2.3****Holomorphic**

If  $f : U \rightarrow \mathbb{C}$  with  $U \subseteq \mathbb{C}$  open, such that  $f$  is differentiable at each point in  $U$ , then  $f$  is *holomorphic*.

**Definition 2.4****Entire**

$f : \mathbb{C} \rightarrow \mathbb{C}$  is called *entire* if it is holomorphic on all of  $\mathbb{C}$ .

**Definition 2.5****Logarithm**

A complex number  $w$  is a *logarithm* of  $z$  if  $e^w = z$ .

**Remark 2.3**

The range of the exponential is  $\mathbb{C} \setminus \{0\}$ . So there is no logarithm of 0.

**Definition 2.6****Branch of Logarithm**

If  $G \subseteq \mathbb{C} \setminus \{0\}$  is a non-empty open set, then a *branch of log* is a continuous function  $\ell : G \rightarrow \mathbb{C}$  such that for all  $z \in G$ ,  $\ell(z)$  is a logarithm of  $z$ , i.e.  $e^{\ell(z)} = z$

**Note**

A branch of log may or may not exist depending on the shape of  $G$ .

**Example 2.1**

$G := \mathbb{C} \setminus (-\infty, 0]$ .  $\ell(z) = \ln|z| + i \operatorname{Arg}(z)$ . Then

$$\begin{aligned} e^{\ell(z)} &= e^{\ln|z|}(\cos(\operatorname{Arg}(z)) + i \sin(\operatorname{Arg}(z))) \\ &= |z|(\cos(\operatorname{Arg}(z)) + i \sin(\operatorname{Arg}(z))) \\ &= z \end{aligned}$$

This is called the *principal branch of log* and is denoted by  $\operatorname{Log}(z)$

**Note**

If we define  $\ell(z) = \text{Log}(z) + i2\pi k$  on  $\mathbb{C} \setminus (-\infty, 0]$  for fixed  $k \in \mathbb{Z}$ , then we get a branch of log.

**Remark 2.4**

It is not possible to find a branch of log on  $\mathbb{C} \setminus \{0\}$ .

**Exercise 2.5**

In polar form, the Cauchy-Riemann equations become, if  $f(z) = u(z) + iv(z)$

$$ru_r = v_\theta$$

$$\text{and } u_\theta = -rv_r$$

**Theorem 2.3****Derivative of Logarithm**

If  $\ell : G \rightarrow \mathbb{C}$  is a branch of log, then  $\ell$  is holomorphic and  $\ell'(z) = \frac{1}{z}$ .

**Theorem 2.4****Polynomial Holomorphicity Criterion**

Suppose  $f$  is a polynomial in variables  $x$  and  $y$  (with complex coefficients), then  $f$  is holomorphic iff  $f$  can be written as polynomial in  $z = x + iy$ .

### 3 Series

**Definition 3.1****Series**

A *series* is a formal sum  $\sum_{n=0}^{\infty} c_n$  where the terms  $c_n \in \mathbb{C}$ . We say a series *converges* if the sequence of partial sums converge otherwise we say it *diverges*.

**Lemma 3.1****Term Test**

If  $\sum_{n=0}^{\infty} c_n$  converges, then  $c_n \rightarrow 0$ .

**Definition 3.2****Absolute Convergence**

A series  $\sum_{n=0}^{\infty} c_n$  *converges absolutely* if  $\sum_{n=0}^{\infty} |c_n| < \infty$

**Lemma 3.2****Absolute Convergence Test**

If  $\sum_{n=0}^{\infty} c_n$  converges absolutely, then  $\sum_{n=0}^{\infty} c_n$  converges and  $|\sum_{n=0}^{\infty} c_n| \leq \sum_{n=0}^{\infty} |c_n|$

**Example 3.1****Geometric Series**

$\sum_{n=0}^{\infty} c^n$  converges iff  $|c| < 1$ . If  $|c| < 1$  then  $\sum_{n=0}^{\infty} c^n = \frac{1}{1-c}$

**Definition 3.3****Convergence of Functions**

Suppose  $(f_n)_{n=1}^{\infty}$  is a sequence of  $\mathbb{C}$ -valued functions on  $G \subseteq \mathbb{C}$ .

- $(f_n)_{n=1}^{\infty}$  *converges pointwise* to  $f$  on  $S \subseteq G$  if  $\lim_{n \rightarrow \infty} f_n(z) = f(z)$  for every  $z \in S$ .
- $(f_n)_{n=1}^{\infty}$  *converges uniformly* to  $f$  on  $S$  if  $\sup_{z \in S} |f_n(z) - f(z)| \rightarrow 0$
- $(f_n)_{n=1}^{\infty}$  *converges locally uniformly* to  $f$  on  $G$  if for each  $z \in G$ , there is some open ball  $B \subseteq G$  with centre  $z$  such that  $(f_n)_{n=1}^{\infty}$  converges uniformly to  $f$  on  $B$ .

**Definition 3.4****Series of Functions**

If  $(f_n)_{n=0}^{\infty}$  is a sequence of functions,  $\sum_{n=0}^{\infty} f_n$  is the associated series of functions. The series *converges* to  $f$  if the partial sums do.

**Example 3.2**

$z_0 \in \mathbb{C}$  is fixed.  $(f_n)_{n=0}^{\infty}$ , where  $f_n(z) = a_n(z - z_0)^n$  we get the series  $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ . The power series centred at  $z_0$ .

**Proposition 3.3****Region of Convergence**

The region of convergence for a power series  $\sum_{n=0}^{\infty} a_n(z - z_0)^n$  is of the form either

1.  $\{z_0\}$
2.  $\mathbb{C}$
3. The points in some disk centred at  $z_0$  and possibly points on the boundary.

**Example 3.3**

$\sum_{n=0}^{\infty} z^n$  has radius of convergence 1. This represents the function  $\frac{1}{1-z}$  on  $\mathbb{D}$ . This converges locally uniformly.

**Theorem 3.4****Comparison Test for Power Series**

Suppose  $\sum_{n=0}^{\infty} a_n(z - z_0)^n$  and  $\sum_{n=0}^{\infty} b_n(z - z_0)^n$  are power series with radius of convergence  $R_1, R_2$  respectively. If  $|b_n| \leq M|a_n|$  for some  $M > 0$ , then  $R_1 \leq R_2$

**Proof:** This follows from the previous proposition because if  $|z - z_0| < R$ , where  $R$  is the radius of convergence, then  $\sum a_n(z - z_0)^n$  converges absolutely.  $\square$

**Theorem 3.5****Cauchy-Hadamard Theorem**

The radius of convergence for a power series  $\sum a_n(z - z_0)^n$  is

$$\frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}}$$

**Theorem 3.6****Differentiation of Power Series**

Suppose  $\sum_{n=0}^{\infty} a_n(z - z_0)^n$  has radius of convergence  $R > 0$ . Let  $f$  be the function it represents (on that disk). Then the radius of convergence for  $\sum_{n=1}^{\infty} n a_n(z - z_0)^{n-1}$  is also  $R$  and if  $g$  is the function it represents, then  $f$  is differentiable and  $f'(z) = g(z)$  for  $z \in G = \{z \in \mathbb{C} \mid |z - z_0| < R\}$

**Proof:** Note

$$\begin{aligned} & \frac{1}{\limsup_{n \rightarrow \infty} |n a_n|^{\frac{1}{n}}} \\ &= \frac{1}{\lim_{n \rightarrow \infty} n^{\frac{1}{n}}} \cdot \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}} \\ &= \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}} \\ &= R \end{aligned}$$

□

**Example 3.4**

$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$  then  $f$  is entire,  $f'(z) = f(z)$  and  $f(0) = 1$ .

**Corollary 3.7****Smoothness of Power Series**

If  $f$  is represented by a power series on its disk of convergence, then derivatives of all orders exist on that disk.

## 4 Complex Integration

**Definition 4.1****Complex Integral**

Suppose  $\varphi : [a, b] \rightarrow \mathbb{C}$  is piecewise continuous (continuous except at finitely many points where one-sided limits exist). Then  $\operatorname{Re} \varphi$  and  $\operatorname{Im} \varphi$  are also piecewise continuous. Set

$$\int_a^b \varphi := \int_a^b \operatorname{Re} \varphi + i \int_a^b \operatorname{Im} \varphi$$

Note this is complex linear in  $\varphi$

**Theorem 4.1****Fundamental Theorem of Calculus**

If  $\varphi$  is continuous and piecewise  $\mathcal{C}^1$  on  $[a, b]$ , then

$$\int_a^b \varphi'(t) dt = \varphi(b) - \varphi(a)$$

**Theorem 4.2****Integral Triangle Inequality**

$$\left| \int_a^b \varphi(t) dt \right| \leq \int_a^b |\varphi(t)| dt$$

**Definition 4.2****Path**

A path is a function  $[a, b] \rightarrow \mathbb{C}$ .

**Definition 4.3****Piecewise  $\mathcal{C}^1$  Path**

A path  $\gamma : [a, b] \rightarrow \mathbb{C}$  is piecewise  $\mathcal{C}^1$  if  $\gamma$  is continuous and there is a partition of  $[a, b]$  such that  $\gamma$  is  $\mathcal{C}^1$  on each subinterval, with one-sided limits for  $\gamma'$  at the breakpoints.

**Definition 4.4****Path Integral**

Suppose  $\gamma : [a, b] \rightarrow \mathbb{C}$  is piecewise- $\mathcal{C}^1$  and  $f : G \rightarrow \mathbb{C}$  is continuous on a set  $G$  containing the image of  $\gamma$ . Then define

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt$$

**Example 4.1**

$f(z) = ie^x$  where  $z = x + iy$ ,  $\gamma : [a, b] \rightarrow \mathbb{C}$ ,  $\gamma(t) = t$  then

$$\int_{\gamma} f(z) dz = \int_a^b ie^t dt = i(e^b - e^a)$$

**Theorem 4.3****Fundamental Theorem for Path Integrals**

If  $\gamma : [a, b] \rightarrow \mathbb{C}$  is piecewise- $\mathcal{C}^1$  and if  $f$  is holomorphic on an open set that contains the range of  $\gamma$  then

$$\int_{\gamma} f'(z) dz = f(\gamma(b)) - f(\gamma(a))$$

**Proof:**

$$\int_{\gamma} f'(z) dz = \int_a^b f'(\gamma(t)) \gamma'(t) dt = \int_a^b (f \circ \gamma)'(t) dt = (f \circ \gamma)(b) - (f \circ \gamma)(a)$$

□

**Proposition 4.4****Reparametrization Invariance**

If  $\gamma : [a, b] \rightarrow \mathbb{C}$  is piecewise- $\mathcal{C}^1$  and  $\beta : [a_1, b_1] \rightarrow [a, b]$  is increasing and piecewise- $\mathcal{C}^1$  with  $\beta(a_1) = a$  and  $\beta(b_1) = b$ . Set  $\gamma_1 = \gamma \circ \beta$ . Suppose  $f$  is continuous on the image of  $\gamma$ . Then

$$\int_{\gamma_1} f(z) \, dz = \int_{\gamma} f(z) \, dz.$$

**Proof:** By definition of the path integral and the chain rule,

$$\begin{aligned} \int_{\gamma_1} f(z) \, dz &= \int_{a_1}^{b_1} f(\gamma_1(s)) \gamma_1'(s) \, ds \\ &= \int_{a_1}^{b_1} f(\gamma(\beta(s))) \gamma'(\beta(s)) \beta'(s) \, ds \end{aligned}$$

By the substitution theorem with  $t = \beta(s)$ ,

$$\begin{aligned} \int_{a_1}^{b_1} f(\gamma(\beta(s))) \gamma'(\beta(s)) \beta'(s) \, ds &= \int_a^b f(\gamma(t)) \gamma'(t) \, dt \\ &= \int_{\gamma} f(z) \, dz \end{aligned}$$

□

**Proposition 4.5****Properties of Path Integrals**

1. If  $a < c < b$  and  $\gamma : [a, b] \rightarrow \mathbb{C}$  is piecewise- $\mathcal{C}^1$ , let  $\gamma_1 = \gamma|_{[a, c]}$  and  $\gamma_2 = \gamma|_{[c, b]}$ . Then

$$\begin{aligned} \int_{\gamma} f(z) \, dz &= \int_a^b f(\gamma(t)) \gamma'(t) \, dt \\ &= \int_a^c f(\gamma(t)) \gamma'(t) \, dt + \int_c^b f(\gamma(t)) \gamma'(t) \, dt \\ &= \int_{\gamma_1} f(z) \, dz + \int_{\gamma_2} f(z) \, dz \end{aligned}$$

2. If  $\gamma : [a, b] \rightarrow \mathbb{C}$  is piecewise- $\mathcal{C}^1$ , define the reverse path  $-\gamma : [a, b] \rightarrow \mathbb{C}$  by  $(-\gamma)(t) = \gamma(a + b - t)$ . Then

$$\begin{aligned} \int_{-\gamma} f(z) \, dz &= \int_a^b f(\gamma(a + b - t)) \cdot (-\gamma'(a + b - t)) \, dt \\ &= - \int_a^b f(\gamma(a + b - t)) \gamma'(a + b - t) \, dt \\ &= - \int_a^b f(\gamma(s)) \gamma'(s) \, ds \\ &= - \int_{\gamma} f(z) \, dz \end{aligned}$$

**Example 4.2**

$z_0 \in \mathbb{C}$ ,  $R > 0$ ,  $n \in \mathbb{Z}$ ,  $\gamma : [0, 2\pi] \rightarrow \mathbb{C}$ ,  $\gamma(t) = z_0 + Re^{it}$ . If  $n \neq -1$ , then

$$\int_{\gamma} (z - z_0)^n dz = 0$$

If  $n = -1$ , then

$$\int_{\gamma} (z - z_0)^n dz = 2\pi i$$

Indeed, if  $n \neq -1$ , then  $F(z) = \frac{1}{n+1}(z - z_0)^{n+1}$  is an antiderivative of  $(z - z_0)^n$ , so

$$\begin{aligned} \int_{\gamma} (z - z_0)^n dz &= F(\gamma(2\pi)) - F(\gamma(0)) \\ &= \frac{1}{n+1}(\gamma(2\pi) - z_0)^{n+1} - \frac{1}{n+1}(\gamma(0) - z_0)^{n+1} \\ &= 0. \end{aligned}$$

If  $n = -1$ , then

$$\begin{aligned} \int_{\gamma} \frac{1}{z - z_0} dz &= \int_0^{2\pi} \frac{1}{\gamma(t) - z_0} \gamma'(t) dt \\ &= \int_0^{2\pi} \frac{1}{Re^{it}} iRe^{it} dt \\ &= \int_0^{2\pi} i dt \\ &= 2\pi i. \end{aligned}$$

**Definition 4.5****Length of a Path**

If  $\gamma : [a, b] \rightarrow \mathbb{C}$  is piecewise- $\mathcal{C}^1$  its length is

$$L(\gamma) := \int_a^b |\gamma'(t)| dt$$

**Theorem 4.6****ML Inequality**

If  $f$  is continuous on the image of  $\gamma$ , then

$$\left| \int_{\gamma} f(z) dz \right| \leq L(\gamma) M_f$$

where  $M_f$  is the maximum value of  $|(f \circ \gamma)(t)|$  for  $t \in [a, b]$ .

**Proof:**

$$\begin{aligned}
\left| \int_{\gamma} f(z) \, dz \right| &= \left| \int_a^b f(\gamma(t)) \gamma'(t) \, dt \right| \\
&\leq \int_a^b |f(\gamma(t)) \gamma'(t)| \, dt \\
&= \int_a^b |f(\gamma(t))| |\gamma'(t)| \, dt \\
&\leq \int_a^b M_f |\gamma'(t)| \, dt \\
&= M_f L(\gamma)
\end{aligned}$$

□

**Theorem 4.7****Uniform Convergence for Path Integrals**

Suppose  $\gamma : [a, b] \rightarrow \mathbb{C}$  is piecewise- $\mathcal{C}^1$  and  $(f_n)_{n=1}^{\infty}$  are continuous on the image of  $\gamma$  and  $(f_n)$  converges uniformly to  $f$  on the range of  $\gamma$ . Then

$$\int_{\gamma} f = \lim_{n \rightarrow \infty} \int_{\gamma} f_n$$

**Proof:** Since  $f_n \rightarrow f$  uniformly and each  $f_n$  is continuous on the image of  $\gamma$ ,  $f$  is continuous on the image of  $\gamma$ . By the ML inequality,

$$\begin{aligned}
\left| \int_{\gamma} f_n - \int_{\gamma} f \right| &= \left| \int_{\gamma} (f_n - f) \right| \\
&\leq L(\gamma) \sup_{t \in [a, b]} |f_n(\gamma(t)) - f(\gamma(t))| \\
&= L(\gamma) \sup_{z \in \gamma([a, b])} |f_n(z) - f(z)|.
\end{aligned}$$

Since  $f_n \rightarrow f$  uniformly on the range of  $\gamma$ , the last expression tends to 0. Hence  $\int_{\gamma} f_n \rightarrow \int_{\gamma} f$ . □

**Example 4.3****Fourier Transform of the Gaussian**

Let  $a, b > 0$  and let  $R_{a,b}$  be the positively oriented boundary of the rectangle with vertices  $-a$ ,  $a$ ,  $a + ib$ , and  $-a + ib$ . Consider

$$\int_{R_{a,b}} e^{-z^2} \, dz.$$

Since

$$e^{-z^2} = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{n!},$$

define

$$g(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)n!}.$$

The power series for  $g$  has infinite radius of convergence, and termwise differentiation gives  $g'(z) = e^{-z^2}$ . Hence

$$\int_{R_{a,b}} e^{-z^2} dz = \int_{R_{a,b}} g'(z) dz = 0.$$

Writing the four sides of  $R_{a,b}$  separately gives

$$0 = \int_{-a}^a e^{-x^2} dx + \int_0^b e^{-(a+iy)^2} i dy - \int_{-a}^a e^{-(x+ib)^2} dx - \int_0^b e^{-(-a+iy)^2} i dy.$$

Thus

$$\int_{-a}^a e^{-(x+ib)^2} dx = \int_{-a}^a e^{-x^2} dx + i \int_0^b e^{-(a+iy)^2} dy - i \int_0^b e^{-(-a+iy)^2} dy.$$

Since

$$e^{-(x+ib)^2} = e^{-x^2+b^2-2ibx},$$

we get

$$\int_{-a}^a e^{-x^2} e^{-2ibx} dx = e^{-b^2} \int_{-a}^a e^{-x^2} dx + i e^{-b^2} \int_0^b e^{-(a+iy)^2} dy - i e^{-b^2} \int_0^b e^{-(-a+iy)^2} dy.$$

For  $0 \leq y \leq b$ ,

$$|e^{-(a+iy)^2}| = e^{-a^2+y^2} \leq e^{-a^2+b^2}$$

and the same bound holds for  $e^{-(-a+iy)^2}$ . Hence the two vertical integrals tend to 0 as  $a \rightarrow \infty$ .

Therefore

$$\int_{-\infty}^{\infty} e^{-x^2} e^{-2ibx} dx = e^{-b^2} \int_{-\infty}^{\infty} e^{-x^2} dx.$$

Since  $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ ,

$$\int_{-\infty}^{\infty} e^{-x^2} e^{-2ibx} dx = \sqrt{\pi} e^{-b^2}.$$

**Theorem 4.8****Cauchy's Theorem for a Triangle (Goursat's Lemma)**

If  $T$  is a triangle in  $\mathbb{C}$  and if  $f$  is holomorphic in an open region that contains  $T$  and its interior, then

$$\int_T f(z) \, dz = 0.$$

**Proof:** Let

$$I = \int_T f(z) \, dz$$

and assume that  $T$  is positively oriented. Divide  $T$  into four congruent subtriangles by joining the midpoints of the sides. The boundary integrals over the four subtriangles add to  $I$ , because the integrals over the shared sides cancel. Hence one subtriangle, call it  $T_1$ , satisfies

$$\left| \int_{T_1} f(z) \, dz \right| \geq \frac{|I|}{4}.$$

Repeating this construction gives nested triangles

$$T \supseteq T_1 \supseteq T_2 \supseteq \dots$$

such that

$$\left| \int_{T_n} f(z) \, dz \right| \geq \frac{|I|}{4^n}.$$

If  $P$  is the perimeter of  $T$  and  $D$  is the diameter of  $T$ , then the perimeter and diameter of  $T_n$  are  $\frac{P}{2^n}$  and  $\frac{D}{2^n}$  respectively.

Since the triangles are nested compact sets and their diameters tend to 0, there is a unique point

$$z_0 \in \bigcap_{n=1}^{\infty} T_n.$$

Since  $f$  is holomorphic at  $z_0$ ,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \eta(z)(z - z_0),$$

where  $\eta(z) \rightarrow 0$  as  $z \rightarrow z_0$ . The first two terms have primitives, so their integrals around  $T_n$  are 0. Thus

$$\int_{T_n} f(z) \, dz = \int_{T_n} \eta(z)(z - z_0) \, dz.$$

Let  $\varepsilon > 0$ . Since  $\eta(z) \rightarrow 0$ , for all sufficiently large  $n$  we have  $|\eta(z)| < \varepsilon$  for all  $z \in T_n$ . Hence, by the ML inequality,

$$\left| \int_{T_n} f(z) dz \right| \leq \varepsilon \cdot \left( \frac{D}{2^n} \right) \cdot \left( \frac{P}{2^n} \right) = \frac{\varepsilon PD}{4^n}.$$

Combining this with the lower bound gives, for all sufficiently large  $n$ ,

$$\frac{|I|}{4^n} \leq \left| \int_{T_n} f(z) dz \right| \leq \frac{\varepsilon PD}{4^n}.$$

Multiplying by  $4^n$ , we get  $|I| \leq \varepsilon PD$ . Since  $\varepsilon > 0$  was arbitrary,  $I = 0$ . □

### Theorem 4.9

### Cauchy's Theorem for Convex Regions

Suppose  $f$  is holomorphic in  $G$  and  $G$  is convex, then for any piecewise- $\mathcal{C}^1$  closed curve  $\gamma$  in  $G$  we have

$$\int_{\gamma} f(z) dz = 0$$

**Proof:** Fix  $z_0 \in G$ . For  $z \in G$ , define

$$F(z) = \int_{[z_0, z]} f(w) dw.$$

This is well-defined since  $G$  is convex. We claim that  $F' = f$ .

Let  $z \in G$ . Since  $G$  is open, for  $h$  sufficiently small we have  $z + h \in G$ . By convexity, the triangle with vertices  $z_0, z, z + h$  is contained in  $G$ , so by Cauchy's theorem for triangles,

$$\int_{[z_0, z]} f(w) dw + \int_{[z, z+h]} f(w) dw - \int_{[z_0, z+h]} f(w) dw = 0.$$

Hence

$$F(z + h) - F(z) = \int_{[z, z+h]} f(w) dw.$$

Parametrizing the line segment  $[z, z + h]$  by  $w = z + th$ ,  $0 \leq t \leq 1$ , gives

$$\frac{F(z + h) - F(z)}{h} = \int_0^1 f(z + th) dt.$$

Therefore

$$\frac{F(z + h) - F(z)}{h} - f(z) = \int_0^1 (f(z + th) - f(z)) dt.$$

Since  $f$  is continuous at  $z$ , the right side tends to 0 as  $h \rightarrow 0$ . Thus  $F'(z) = f(z)$ .

Now if  $\gamma : [a, b] \rightarrow G$  is a piecewise- $\mathcal{C}^1$  closed curve, then by [Theorem 4.3](#),

$$\int_{\gamma} f(z) dz = F(\gamma(b)) - F(\gamma(a)) = 0,$$

since  $\gamma(a) = \gamma(b)$ . □

### Theorem 4.10

### Cauchy's Formula for the Circle

Suppose  $C$  is a circle oriented counter-clockwise and  $f$  is holomorphic in a region that contains  $C$  and its interior, then if  $z$  is in the interior of  $C$

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta$$

**Proof:** Let

$$g(\zeta) = \begin{cases} \frac{f(\zeta) - f(z)}{\zeta - z} & \text{if } \zeta \neq z \\ f'(z) & \text{if } \zeta = z. \end{cases}$$

Then  $g$  is holomorphic in the interior of  $C$ , so by Cauchy's theorem for convex regions,

$$\int_C g(\zeta) d\zeta = 0.$$

Hence

$$\int_C \frac{f(\zeta)}{\zeta - z} d\zeta = f(z) \int_C \frac{1}{\zeta - z} d\zeta.$$

It remains to compute the last integral. Write  $C(t) = z_0 + Re^{it}$ ,  $0 \leq t \leq 2\pi$ , and set  $a = \frac{z - z_0}{R}$ . Since  $z$  is in the interior of  $C$ ,  $|a| < 1$ . Then

$$\begin{aligned} \int_C \frac{1}{\zeta - z} d\zeta &= \int_0^{2\pi} \frac{iRe^{it}}{z_0 + Re^{it} - z} dt \\ &= \int_0^{2\pi} \frac{i}{1 - ae^{-it}} dt \\ &= \int_0^{2\pi} i \sum_{n=0}^{\infty} a^n e^{-int} dt \\ &= 2\pi i. \end{aligned}$$

The third equality uses the geometric series expansion, valid uniformly since  $|a| < 1$ . The last equality follows by integrating the uniformly convergent series term-by-term: all terms with  $n \geq 1$  integrate to 0, and the  $n = 0$  term gives  $2\pi i$ . Therefore

$$\int_C \frac{f(\zeta)}{\zeta - z} d\zeta = 2\pi i f(z),$$

which gives the result. □

**Note**

The integral is 0 for  $z$  outside the circle

**Corollary 4.11****Mean Value Property**

Suppose  $f$  is holomorphic on a region containing  $\{z \in \mathbb{C} \mid |z - z_0| < R\}$ . If  $0 < r < R$ , then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

**Proof:** Let  $C_r$  be the circle of radius  $r$  centered at  $z_0$ , oriented counter-clockwise. By Cauchy's formula,

$$f(z_0) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{\zeta - z_0} d\zeta.$$

Parametrize  $C_r$  by  $\gamma(t) = z_0 + re^{it}$  for  $0 \leq t \leq 2\pi$ . Then  $\gamma'(t) = ire^{it}$ , so

$$\begin{aligned} f(z_0) &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{it})}{re^{it}} ire^{it} dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt. \end{aligned}$$

□

**Note**

This is called the *mean value property* because  $f(z_0)$  is equal to the average of the values of  $f$  on the circle centered at  $z_0$ .

**Definition 4.6**

If  $\varphi$  is a continuous function on the range of some piecewise- $\mathcal{C}^1$  path  $\gamma$ . The *Cauchy Integral* over  $\gamma$  is the function

$$z \mapsto \int_{\gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta$$

This is defined for  $z$  not in the range of  $\gamma$

**Proposition 4.12****Power Series of a Cauchy Integral**

Let  $\gamma$  be a piecewise- $\mathcal{C}^1$  path and suppose  $\varphi$  is continuous on the range of  $\gamma$ . Define

$$f(z) = \int_{\gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta$$

for  $z$  not in the range of  $\gamma$ . If  $z_0$  is not in the range of  $\gamma$  and

$$R = \text{dist}(z_0, \text{range}(\gamma)),$$

then  $f$  has a power series representation on  $\{z \in \mathbb{C} \mid |z - z_0| < R\}$ .

**Proof:** Let  $z$  satisfy  $|z - z_0| < R$ . Then for  $\zeta$  in the range of  $\gamma$ ,

$$\left| \frac{z - z_0}{\zeta - z_0} \right| < 1.$$

Thus

$$\begin{aligned} \frac{1}{\zeta - z} &= \frac{1}{\zeta - z_0 - (z - z_0)} \\ &= \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} \\ &= \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}}. \end{aligned}$$

Therefore

$$f(z) = \int_{\gamma} \varphi(\zeta) \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}} d\zeta.$$

If  $0 < \rho < R$  and  $|z - z_0| \leq \rho$ , then

$$\left| \frac{z - z_0}{\zeta - z_0} \right| \leq \frac{\rho}{R} < 1$$

for all  $\zeta$  in the range of  $\gamma$ . Hence the geometric series converges uniformly on  $|z - z_0| \leq \rho$  and on the range of  $\gamma$ , so we may integrate term by term. Thus

$$f(z) = \sum_{n=0}^{\infty} \left( \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \right) (z - z_0)^n.$$

This is a power series centered at  $z_0$ . □

**Theorem 4.13****Holomorphic Functions are Analytic**

Suppose  $f$  is holomorphic on a region containing  $\{z \in \mathbb{C} \mid |z - z_0| < R\}$ . Then  $f$  has a power series representation centered at  $z_0$  with radius of convergence at least  $R$ . More precisely, for  $0 < r < R$  and  $|z - z_0| < r$ ,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

and  $C_r$  is the positively oriented circle of radius  $r$  centered at  $z_0$ .

**Proof:** Fix  $0 < r < R$ . By Cauchy's formula, for  $|z - z_0| < r$ ,

$$f(z) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

This is a Cauchy integral with  $\varphi(\zeta) = \frac{f(\zeta)}{2\pi i}$ . By the power series representation for Cauchy integrals,

$$f(z) = \sum_{n=0}^{\infty} \left( \int_{C_r} \frac{f(\zeta)}{2\pi i (\zeta - z_0)^{n+1}} d\zeta \right) (z - z_0)^n.$$

Hence

$$a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Since  $r < R$  was arbitrary, the radius of convergence is at least  $R$ . □

**Corollary 4.14****Cauchy's Integral Formula for Derivatives**

Suppose  $f$  is holomorphic on a region containing  $\{z \in \mathbb{C} \mid |z - z_0| < R\}$ . If  $0 < r < R$  and  $C_r$  is the positively oriented circle of radius  $r$  centered at  $z_0$ , then for every  $n \geq 0$ ,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

**Proof:** By the previous theorem, for  $|z - z_0| < r$ ,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Since power series can be differentiated term-by-term inside their radius of convergence,

$$f^{(n)}(z_0) = n!a_n.$$

Substituting the formula for  $a_n$  gives

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

□

### Corollary 4.15

### Holomorphic Derivatives

If  $f$  is holomorphic on a region  $G$ , then  $f^{(n)}$  is holomorphic on  $G$  for every  $n \geq 0$ .

**Proof:** Let  $z_0 \in G$ . Since  $G$  is open, there is  $R > 0$  such that  $\{z \in \mathbb{C} \mid |z - z_0| < R\} \subseteq G$ . By the theorem above,  $f$  has a power series representation on this disk. Power series can be differentiated term-by-term inside their radius of convergence, so  $f'$  is holomorphic on the disk. Repeating this argument gives that  $f^{(n)}$  is holomorphic near  $z_0$  for every  $n \geq 0$ . Since  $z_0 \in G$  was arbitrary,  $f^{(n)}$  is holomorphic on  $G$ . □

### Proposition 4.16

### Cauchy Product of Power Series

Suppose  $\sum_{n=0}^{\infty} a_n(z - z_0)^n$  has radius of convergence  $R_1$  and  $\sum_{n=0}^{\infty} b_n(z - z_0)^n$  has radius of convergence  $R_2$ . Consider

$$\sum_{n=0}^{\infty} \left( \sum_{k=0}^n a_k b_{n-k} \right) (z - z_0)^n$$

Then the radius of convergence for this power series is at least  $\min\{R_1, R_2\}$ . If  $f$  and  $g$  are the functions defined by the first two power series, then for  $|z - z_0| < \min\{R_1, R_2\}$ ,

$$f(z)g(z) = \sum_{n=0}^{\infty} \left( \sum_{k=0}^n a_k b_{n-k} \right) (z - z_0)^n.$$

**Proof:** Let  $R = \min\{R_1, R_2\}$  and define  $h(z) = f(z)g(z)$  for  $|z - z_0| < R$ . Since  $f$  and  $g$  are holomorphic on this disk, so is  $h$ . Thus  $h$  has a power series representation

$$h(z) = \sum_{n=0}^{\infty} c_n(z - z_0)^n$$

on  $|z - z_0| < R$ , where  $c_n = h^{(n)}(z_0)/n!$ . By Leibniz's rule,

$$\begin{aligned}
h^{(n)}(z_0) &= \sum_{k=0}^n \binom{n}{k} f^{(k)}(z_0) g^{(n-k)}(z_0) \\
&= \sum_{k=0}^n \frac{n!}{k!(n-k)!} \cdot k! a_k \cdot (n-k)! b_{n-k} \\
&= n! \sum_{k=0}^n a_k b_{n-k}.
\end{aligned}$$

Hence

$$c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Therefore the Cauchy product represents  $h = fg$  on  $|z - z_0| < R$ , so its radius of convergence is at least  $R$ . □

### Proposition 4.17

### Converse of Goursat's Lemma

If  $f : G \rightarrow \mathbb{C}$  is continuous and  $\int_T f = 0$  for every triangle  $T$  whose edges and interior are contained in  $G$ , then  $f$  is holomorphic.

**Proof:** It is enough to show that  $f$  is holomorphic in a neighbourhood of each point. Fix  $z_0 \in G$  and choose a disk  $D$  centered at  $z_0$  with  $D \subseteq G$ .

For  $z \in D$ , define

$$g(z) = \int_{[z_0, z]} f(\zeta) d\zeta.$$

The triangle hypothesis implies that this integral is path independent in  $D$ : if  $z_1, z_2, z_3 \in D$ , then the integral around the triangle with these vertices is 0.

By the same line-segment argument as in the proof of Cauchy's theorem for convex regions,  $g' = f$  on  $D$ . Therefore  $g$  is holomorphic on  $D$ . Since holomorphic derivatives are holomorphic,  $f = g'$  is holomorphic on  $D$ . Since  $z_0$  was arbitrary,  $f$  is holomorphic on  $G$ . □

### Theorem 4.18

### Liouville's Theorem

Suppose  $f : \mathbb{C} \rightarrow \mathbb{C}$  is entire and bounded. Then  $f$  is constant.

**Proof:** Suppose  $|f(z)| \leq M$  for all  $z \in \mathbb{C}$ . Fix  $z_0 \in \mathbb{C}$  and  $r > 0$ . Let  $C_r$  be the circle centered at  $z_0$  with radius  $r$ , oriented counter-clockwise. By Cauchy's integral formula for derivatives,

$$f'(z_0) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^2} d\zeta.$$

By the ML inequality,

$$|f'(z_0)| \leq \frac{1}{2\pi} \cdot \left( \frac{M}{r^2} \right) \cdot 2\pi r = \frac{M}{r}.$$

Since this holds for every  $r > 0$ , letting  $r \rightarrow \infty$  gives  $f'(z_0) = 0$ . Since  $z_0$  was arbitrary,  $f' = 0$  on  $\mathbb{C}$ , so  $f$  is constant.  $\square$

### Theorem 4.19

### The Fundamental Theorem of Algebra

If  $p(z) = \sum_{k=0}^n a_k z^k$  is a nonconstant polynomial with complex coefficients and  $a_n \neq 0$ , then there exist  $z_1, \dots, z_n \in \mathbb{C}$  such that

$$p(z) = a_n \prod_{k=1}^n (z - z_k).$$

**Proof:** It is enough to show that  $p$  has a zero, since after finding a zero  $z_1$ , we may factor  $p(z) = (z - z_1)q(z)$  and repeat by induction on the degree.

Suppose, for contradiction, that  $p$  has no zero. Then  $\frac{1}{p}$  is entire. We claim that  $\frac{1}{p}$  is bounded. Since  $a_n \neq 0$ ,

$$p(z) = z^n \left( a_n + \frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right).$$

For  $|z|$  sufficiently large, the term in parentheses has absolute value at least  $\frac{|a_n|}{2}$ , so

$$|p(z)| \geq \frac{|a_n|}{2} \cdot |z|^n.$$

Hence  $\left| \frac{1}{p(z)} \right| \rightarrow 0$  as  $|z| \rightarrow \infty$ , so  $\frac{1}{p}$  is bounded outside a large disk. On the closed disk,  $\frac{1}{p}$  is continuous, hence bounded. Thus  $\frac{1}{p}$  is bounded on  $\mathbb{C}$ .

By Liouville's theorem,  $\frac{1}{p}$  is constant, so  $p$  is constant, contradicting that  $p$  is nonconstant. Therefore  $p$  has a zero, and the factorization follows by induction.  $\square$

## 5 Zeros

### Definition 5.1

### Zero of Order $n$

Suppose  $f$  is holomorphic on  $G$  and  $f(z_0) = 0$ . We say  $z_0$  is a *zero of order  $n$*  if

$$0 = f(z_0) = f'(z_0) = \dots = f^{(n-1)}(z_0)$$

but  $f^{(n)}(z_0) \neq 0$ . A zero of order 1 is called a *simple zero*.

### Proposition 5.1

### Infinite-Order Zero

Suppose  $G$  is open,  $f$  is holomorphic on  $G$ , and  $f$  has a zero of infinite order at  $z_0 \in G$ . Then  $f = 0$  on the connected component of  $G$  containing  $z_0$ .

**Proof:** Let  $H$  be the connected component of  $G$  containing  $z_0$ . Define

$$A = \{z \in H \mid f \text{ is identically zero on some disk centered at } z\}.$$

Since  $f$  has a zero of infinite order at  $z_0$ , every coefficient in the power series for  $f$  centered at  $z_0$  is 0. Thus  $f$  is identically zero on some disk centered at  $z_0$ , so  $z_0 \in A$  and  $A$  is non-empty.

The set  $A$  is open in  $H$  by definition. We show that  $A$  is closed in  $H$ . Suppose  $z_k \in A$  and  $z_k \rightarrow z \in H$ . For each  $m \geq 0$ , since  $f$  is identically zero near each  $z_k$ , we have

$$f^{(m)}(z_k) = 0.$$

Since  $f^{(m)}$  is continuous,  $f^{(m)}(z) = 0$ . Hence all derivatives of  $f$  vanish at  $z$ , so the power series for  $f$  centered at  $z$  is identically zero on some disk centered at  $z$ . Thus  $z \in A$ .

Therefore  $A$  is non-empty, open, and closed in  $H$ . Since  $H$  is connected,  $A = H$ , so  $f = 0$  on  $H$ .  $\square$

### Proposition 5.2

### Finite-Order Zeros are Isolated

Suppose  $f$  is holomorphic on an open set  $G$  and  $z_0$  is a zero of finite order  $n$ . Then there is a disk centered at  $z_0$  on which  $z_0$  is the only zero of  $f$ .

**Proof:** Since  $z_0$  is a zero of order  $n$ , the power series for  $f$  centered at  $z_0$  has the form

$$f(z) = \sum_{k=n}^{\infty} a_k (z - z_0)^k$$

with  $a_n \neq 0$ . Thus

$$f(z) = (z - z_0)^n g(z),$$

where

$$g(z) = \sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k.$$

Then  $g$  is holomorphic near  $z_0$  and  $g(z_0) = a_n \neq 0$ . By continuity,  $g(z) \neq 0$  on some disk centered at  $z_0$ . On this disk, the only zero of  $f(z) = (z - z_0)^n g(z)$  is  $z_0$ .  $\square$

### Corollary 5.3

### Identity Theorem

Suppose  $f$  and  $g$  are holomorphic on an open set  $G$ . If  $f$  and  $g$  agree on a subset  $S \subseteq G$  that has a limit point  $z_0 \in G$ , then  $f = g$  on the connected component of  $G$  containing  $z_0$ .

**Proof:** Let  $h = f - g$ . Then  $h$  is holomorphic on  $G$  and  $h = 0$  on  $S$ . Since  $z_0$  is a limit point of  $S$ , there is a sequence  $(z_n)$  in  $S$  with  $z_n \rightarrow z_0$  and  $z_n \neq z_0$ . Since  $h$  is continuous,  $h(z_0) = 0$ .

If  $h$  had a zero of finite order at  $z_0$ , then  $z_0$  would be an isolated zero of  $h$ , contradicting that  $z_n \rightarrow z_0$  with  $z_n \neq z_0$  and  $h(z_n) = 0$ . Hence  $h$  has a zero of infinite order at  $z_0$ . By the infinite-order zero proposition,  $h = 0$  on the connected component of  $G$  containing  $z_0$ . Thus  $f = g$  there.  $\square$

### Example 5.1

Suppose  $f : \mathbb{D} \rightarrow \mathbb{C}$  is holomorphic and  $f(\frac{1}{n}) = \frac{1}{n^2}$  for all  $n \in \mathbb{N}$ . What is  $f'(\frac{i}{2})$ ?

**Proof:** Let  $g(z) = z^2$ . Then  $f(\frac{1}{n}) = g(\frac{1}{n})$  for all  $n \in \mathbb{N}$ , and  $\frac{1}{n} \rightarrow 0 \in \mathbb{D}$ . By the identity theorem,  $f = g$  on  $\mathbb{D}$ . Hence

$$f'\left(\frac{i}{2}\right) = g'\left(\frac{i}{2}\right) = 2 \cdot \frac{i}{2} = i.$$

□

### Proposition 5.4

### Weierstrass Theorem

Suppose  $G \subseteq \mathbb{C}$  is open and  $f_k : G \rightarrow \mathbb{C}$  is holomorphic for each  $k \in \mathbb{N}$ . If  $f_k \rightarrow f$  locally uniformly on  $G$ , then  $f$  is holomorphic. Moreover, for each  $n \geq 0$ ,  $f_k^{(n)} \rightarrow f^{(n)}$  locally uniformly on  $G$ .

**Proof:** Fix  $z_0 \in G$  and choose  $R > 0$  such that the closed disk  $\overline{B(z_0, R)} \subseteq G$ . Let  $C$  be the positively oriented circle centered at  $z_0$  with radius  $R$ .

If  $z \in B(z_0, R)$ , then by Cauchy's formula,

$$f_k(z) = \frac{1}{2\pi i} \int_C f_k \frac{\zeta}{\zeta - z} d\zeta.$$

Since  $f_k \rightarrow f$  uniformly on  $C$ , the uniform convergence theorem for path integrals gives

$$f(z) = \lim_{k \rightarrow \infty} f_k(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Thus  $f$  is a Cauchy integral on  $B(z_0, R)$ , so  $f$  is holomorphic there. Since  $z_0$  was arbitrary,  $f$  is holomorphic on  $G$ .

If  $0 < r < R$  and  $z \in B(z_0, r)$ , then by Cauchy's integral formula for derivatives,

$$f_k^{(n)}(z) - f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f_k(\zeta) - f(\zeta)}{(\zeta - z)^{n+1}} d\zeta.$$

Since  $|\zeta - z| \geq R - r$  for  $\zeta \in C$  and  $z \in B(z_0, r)$ , the ML inequality gives

$$\left| f_k^{(n)}(z) - f^{(n)}(z) \right| \leq n! \frac{R}{(R - r)^{n+1}} \sup_{\zeta \in C} |f_k(\zeta) - f(\zeta)|.$$

The supremum tends to 0 by local uniform convergence, so  $f_k^{(n)} \rightarrow f^{(n)}$  uniformly on  $B(z_0, r)$ . Since  $z_0$  was arbitrary, the convergence is locally uniform on  $G$ . □

### Theorem 5.5

### Maximum Modulus Principle

If  $G \subseteq \mathbb{C}$  is a region and  $f : G \rightarrow \mathbb{C}$  is holomorphic and nonconstant, then  $|f|$  has no local maximum in  $G$ .

**Proof:** Suppose  $|f|$  has a local maximum at  $z_0 \in G$ . Choose  $R > 0$  such that  $\{z \in \mathbb{C} \mid |z - z_0| < R\} \subseteq G$  and

$$|f(z)| \leq |f(z_0)|$$

for all  $|z - z_0| < R$ .

If  $f(z_0) = 0$ , then  $|f(z)| = 0$  on this disk, so  $f$  is locally constant. By the identity theorem,  $f$  is constant on  $G$ , a contradiction. Thus  $f(z_0) \neq 0$ .

For  $0 < r < R$ , the mean value property gives

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

Taking absolute values,

$$|f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})| dt \leq |f(z_0)|.$$

Hence equality holds throughout. Since the continuous function

$$t \mapsto |f(z_0)| - |f(z_0 + re^{it})|$$

is nonnegative and has integral 0, it is identically 0. Thus  $|f(z_0 + re^{it})| = |f(z_0)|$  for all  $t$ .

Equality in the triangle inequality also forces all values  $f(z_0 + re^{it})$  to have the same argument as  $f(z_0)$ . Therefore

$$f(z_0 + re^{it}) = f(z_0)$$

for all  $t$ . Since this holds for every  $0 < r < R$ ,  $f = f(z_0)$  on the disk centered at  $z_0$  with radius  $R$ . By the identity theorem,  $f$  is constant on  $G$ , contradicting the hypothesis. Therefore  $|f|$  has no local maximum in  $G$ . □

### Corollary 5.6

If  $f : G \rightarrow \mathbb{C}$  is holomorphic, nonconstant and  $K \subseteq G$  compact and nonempty, then  $|f|$  attains its maximum on  $\partial K$ .

### Theorem 5.7

### Schwarz's Lemma

Suppose  $f : \mathbb{D} \rightarrow \mathbb{D}$  is holomorphic and  $f(0) = 0$ . Then  $|f(z)| \leq |z|$  for all  $z \in \mathbb{D}$ . Moreover, if  $|f(z_0)| = |z_0|$  for some  $z_0 \in \mathbb{D}$ ,  $z_0 \neq 0$ , then there exists  $\lambda \in \mathbb{C}$  with  $|\lambda| = 1$  such that  $f(z) = \lambda z$  for all  $z \in \mathbb{D}$ .

**Proof:** Define

$$g(z) = \begin{cases} \frac{f(z)}{z} & \text{if } z \neq 0 \\ f'(0) & \text{if } z = 0. \end{cases}$$

Since  $f(0) = 0$ , the singularity at 0 is removable, so  $g$  is holomorphic on  $\mathbb{D}$ .

Fix  $0 < r < 1$ . On the circle  $|z| = r$ ,

$$|g(z)| = \frac{|f(z)|}{|z|} < \frac{1}{r}.$$

By the maximum modulus principle, for  $|z| \leq r$  we have  $|g(z)| \leq \frac{1}{r}$ . Letting  $r \rightarrow 1$  gives  $|g(z)| \leq 1$  for all  $z \in \mathbb{D}$ . Hence, for  $z \neq 0$ ,

$$|f(z)| = |z||g(z)| \leq |z|,$$

and the inequality is also true at  $z = 0$ .

Now suppose  $|f(z_0)| = |z_0|$  for some  $z_0 \neq 0$ . Then  $|g(z_0)| = 1$ , so  $|g|$  has a local maximum in  $\mathbb{D}$ . By the maximum modulus principle,  $g$  is constant. Thus  $g(z) = \lambda$  for some  $\lambda \in \mathbb{C}$  with  $|\lambda| = 1$ , and so  $f(z) = \lambda z$  for all  $z \in \mathbb{D}$ . □